THE DIALECT GAP: MODEL RECENCY DICTATES COMMONSENSE REASONING PERFORMANCE IN SOUTH SLAVIC VARIETIES

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1 INTRODUCTION

Most current large language models perform well on standard South Slavic languages, but their usability drops sharply on dialects: varieties that appear rarely in training data and are almost absent from benchmarks, yet are used daily by millions of speakers. Whether modern LLMs can handle dialectal input is therefore an open and practically important question, and one that standard benchmarks do not answer. The South Slavic language group is a well-suited setting for investigating it: it includes several standard languages (Slovenian, Croatian, Serbian, Macedonian) alongside living dialects with distinct phonology, lexicon, and morphology, including Chakavian, the Slovenian Cerkno dialect, Serbian Torlak, and Slovenian Prlekija.

Our prior work (Pungeršek et al., 2025), benchmarked zero-shot LLM prompting on classification and commonsense reasoning across South Slavic languages, showing that frontier models match or approach fine-tuned baselines on the standard languages. That standard-language result is inherited here and is not our contribution. The present paper asks what standard-language evaluation cannot show: how models behave on dialect input, and whether the dialect gap is closing across model generations. Our contribution is threefold. First, we extend the model pool with the newest frontier releases, enabling a comparison across three model generations under one protocol, with earlier results reproduced from our prior work under an identical protocol, test set, and scoring procedure. Second, we operationalize the dialect gap as the difference between each model's mean standard-language and mean dialect accuracy, and track this quantity across generations. Third, we report confidence intervals and paired significance tests for all headline comparisons.

2 BENCHMARKS AND EXPERIMENTAL SETUP

The evaluation protocol, prompt templates, label definitions, dataset splits, and scoring procedure follow our prior work. We evaluate on two benchmarks:

- Global PIQA: a South Slavic subset of Global Physical Interaction Question Answering (Chang et al., 2025), covering English, Slovenian, Croatian, Serbian (Latin and Cyrillic scripts), Macedonian, and the dialects Cerkno, Chakavian, Torlak (both scripts), and Prlekija
- COPA: a causal commonsense suite extending DIALECT-COPA (Ljubešić et al., 2024a), covering English, Slovenian, Croatian, Serbian, and Macedonian plus the dialects Cerkno, Chakavian, Torlak, and Prlekija, with 500 test items per language.

Table 1 lists all models:

Models evaluated on PIQA dataset	Models evaluated on COPA dataset
GPT 5.4 Pro	GPT 5.4 Pro
GPT 5.4	GPT 5.4
GPT 5	GPT 5
GPT 4o	GPT 4o
GPT 3.5 Turbo	GPT 3.5 Turbo
Gemini 3.1 Pro	Gemini 3.1 Pro
Gemini 3.1 Flash Lite	Gemini 3.1 Flash Lite
Gemini 2.5 Pro	Gemini 2.5 Pro
Gemini 2.5 Flash	Gemini 2.5 Flash
Gemma 4 31B IT	Gemma 4 31B IT
Gemma 4 26B A4B IT	Gemma 4 26B A4B IT
Claude Opus 4.6	Claude Opus 4.6
Claude Sonnet 4.6	Claude Sonnet 4.6
Claude Haiku 4.5	Claude Haiku 4.5
Llama 4 Maverick	Llama 4 Maverick
Llama 3.3 70B	Llama 3.3 70B
Mistral Large	Mistral Large
Mistral Medium	Mistral Medium
Mistral Small	Mistral Small
Qwen 3 32B	Qwen 3 32B

Table 1: Models evaluated on the PIQA and COPA datasets.

All models are queried through the OpenRouter API with a retry schedule of 30, 60, 90, 120, and 180 seconds on rate-limit errors. Using a single unified access stack removes one source of variance between conditions at the cost of routing all traffic through a third-party gateway. All results are single-run. We mitigate the resulting uncertainty statistically rather than by repetition: because every model answers the identical items, headline pairwise comparisons use McNemar's exact test on paired outcomes, and we report Wilson 95% confidence intervals for accuracies. At n = 100 items (PIQA), the interval half-width is roughly ±8–10 points in the observed accuracy range, so PIQA differences below 10 points are not interpreted; at n = 500 (COPA) it is roughly ±2–4 points.

We evaluate all models by direct zero-shot classification. Each model receives the task in the source language with both candidate options and selects the more plausible one, returning a JSON object with a binary answer. Both tasks are binary forced-choice with label-balanced test sets, so random guessing converges to 0.5 accuracy; scores clustering around 0.5 indicate the model is effectively guessing rather than reasoning. Predicted labels are extracted in two stages: strict JSON parsing, then a regular-expression fallback when reasoning text surrounds the JSON, a pattern most common with Claude variants. If both stages fail, or the API returns an empty or errored response after the retry schedule, the item is scored incorrect. Reported accuracy is therefore *effective* accuracy under our prompting protocol.

The DIALECT-COPA training and validation splits have been publicly available for two years. The test split we evaluate was not public at the time each model's training data was collected. Global PIQA was released in 2025; models with earlier training cutoffs cannot have seen it, whereas for the newest systems contamination cannot be excluded. Two observations argue against contamination driving our headline result. The same newest models that excel on COPA dialects still fail to close PIQA-Chakavian. We nevertheless treat cross-generation comparisons, run under one protocol, as the primary evidence rather than absolute scores.

3 RESULTS

Figures 2 and 1 summarize accuracy across all conditions for COPA and PIQA.

Frontier models perform at or near English parity on Slovenian, Croatian, Serbian, and Macedonian on both benchmarks, with the strongest systems at or above 0.95 throughout and differences within confidence intervals. Serbian script choice shows no consistent effect: Cyrillic and Latin conditions produce essentially identical scores for every model that handles either

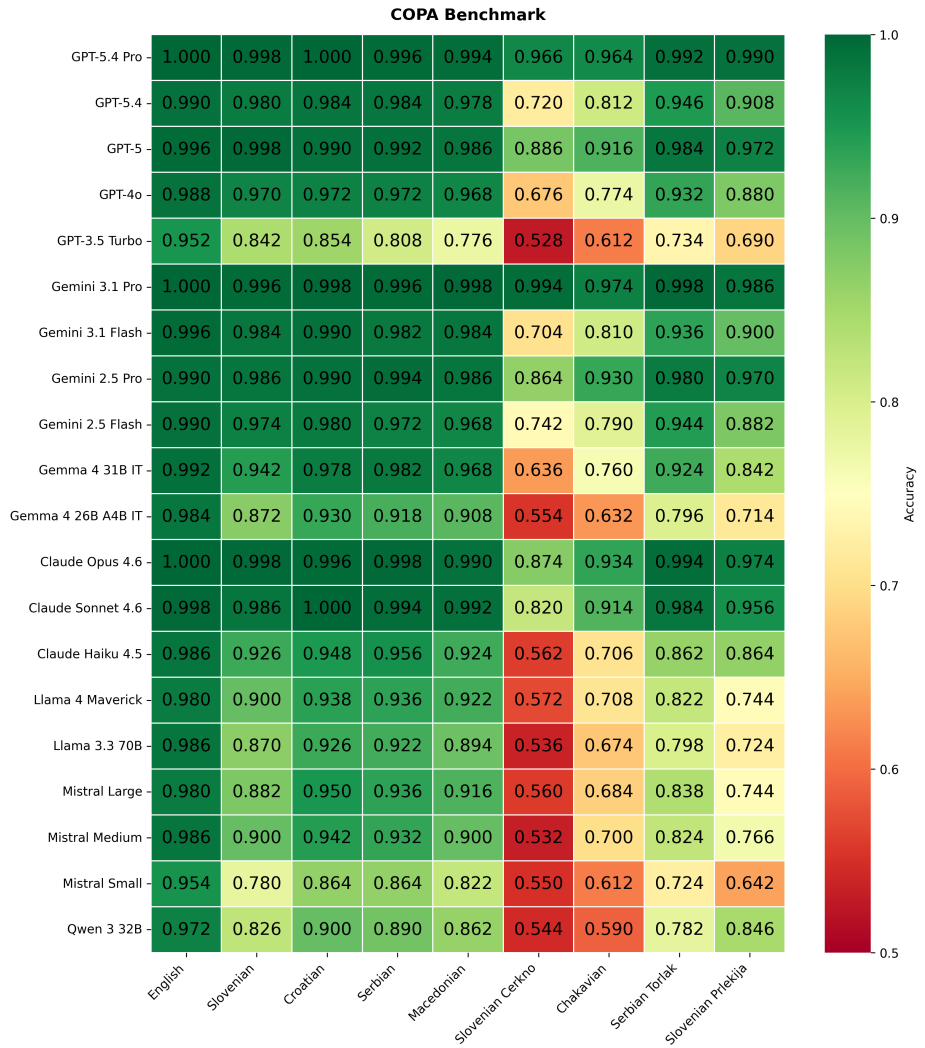


Figure 1: Accuracy of all models on the COPA dataset across all languages.

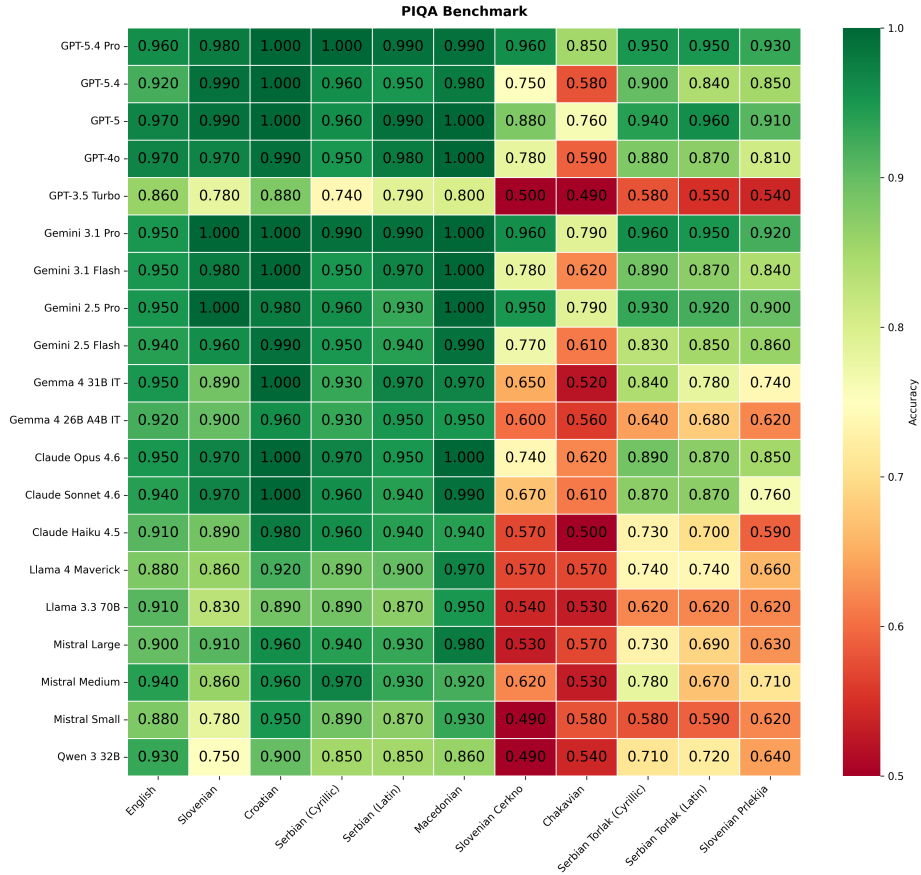


Figure 2: Accuracy of all models on the PIQA dataset across all languages.

Most models lose 20–40 accuracy points moving from standard varieties to Chakavian or Cerkno, and several mid-tier systems collapse to near-chance. On PIQA, the Chakavian drop is consistently the most severe: models that handle Cerkno reasonably still fall toward chance on Chakavian.

Table 2 shows shrinking across release generations, and the effect is cleanest *within* provider families, where architecture lineage and data pipeline are most comparable — the comparison R2 identified as our strongest evidence [CLAIM RESIZED: "recency dictates performance" → "the gap narrows with generation; generation is the strongest observable correlate in our pool"]. On COPA, within the OpenAI flagship line, falls monotonically across four generations: 0.179 (GPT-3.5 Turbo) → 0.155 (GPT-4o) → 0.052 (GPT-5) → 0.019 (GPT-5.4 Pro). The Gemini flagship line shows

the same movement (0.053 for 2.5 Pro → 0.009 for 3.1 Pro). [All values derived from Fig. 1 as printed — AUTHOR: recompute from final run.] Across the full pool, release date correlates with at Spearman = [AUTHOR], p = [AUTHOR], and with mean dialect accuracy at = [AUTHOR]. Parameter count is a weaker predictor within the open-weight subset where it is observable (= [AUTHOR]); for closed models parameter counts are undisclosed, so we make no scale claims about them. An important qualification: the generational effect concentrates in flagship tiers. Lighter same-generation variants show gaps comparable to previous-generation flagships (COPA : GPT-5.4 at 0.135 vs. GPT-5 at 0.052; Gemini 3.1 Flash at 0.148 vs. Gemini 2.5 Pro at 0.053 [derived from Fig. 1 — AUTHOR: verify]), and within a single generation the tier gradient is large (Claude Haiku 4.5 / Sonnet 4.6 / Opus 4.6: = 0.190 / 0.075 / 0.052). Generation shifts the frontier; it does not lift all tiers equally.

4 DISCUSSION

Dialectal items are not simply harder due to regional customs or local knowledge. Two observations support this. First, the strongest current systems answer near-perfectly the same COPA dialect items on which other frontier models fail, so the items are solvable. Second, the degradation appears on COPA, which tests abstract causal reasoning, as strongly as on PIQA, which is grounded in physical and culturally embedded situations. If cultural specificity alone were responsible, models would struggle on PIQA dialects but not on COPA dialects; their joint failure points to the dialectal form as the primary source of difficulty.

he most plausible driver of the generational trend is expanded multilingual — and incidentally dialectal — pretraining coverage, consistent with (Ljubešić et al., 2024b), who showed that even modest in-context dialect exposure improves mid-tier models. We stress that this is a hypothesis, not a measured cause: "recency" bundles training-data cutoff and coverage, post-training methods, and architecture, which our observational design cannot separate. Within-family comparisons hold the provider constant and still show the gap closing, which weakens purely architectural explanations, but the underlying mechanism remains unidentified.

The persistence of the PIQA-Chakavian gap even for the strongest systems is the main open puzzle. We hypothesize two contributing factors. Chakavian is more lexically distant from standard Croatian than Cerkno is from standard Slovenian, and PIQA items are lexically dense: they hinge on specific nouns for tools, containers, food items, and everyday objects, precisely the vocabulary most likely to diverge across varieties — whereas COPA items rely on abstract causal frames that survive lexical

substitution. Aggregate accuracy cannot separate lexical from cultural sources of difficulty, so we treat this account as a hypothesis and make it the primary target of the item-level analysis planned below.

Our design is single-run at requested temperature 0, with the temperature caveat above; test sets are small for PIQA (CIs reported); contamination cannot be fully excluded for the newest models (Section 2); parsing failures are scored incorrect and quantified in Appendix; and routing all queries through one gateway means provider-side serving changes are outside our control.

5 CONCLUSION

Frontier LLMs have reached parity with English on standard South Slavic commonsense reasoning, but dialect input reveals a gap invisible in standard evaluation: most models lose 20–40 points on Chakavian and Cerkno. The gap narrows sharply with model generation at the flagship tier — the newest systems approach standard-language parity on all COPA dialects, while PIQA-Chakavian remains unsolved at, and lighter model tiers lag roughly a generation behind.

Future work follows three technical directions [reframed per R3 — continuous re-benchmarking demoted from headline direction to infrastructure note]. First, item-level error analysis on PIQA-Chakavian to separate lexical from cultural sources of difficulty, including error-overlap analysis across model generations. Second, few-shot prompting with in-context dialectal examples as a tractable, retraining-free intervention for mid-tier open-weight models, following (Ljubešić et al., 2024b). Third, extending dialect coverage to further low-resource varieties. We maintain the benchmark for re-evaluation as new models appear.

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A GLOBAL PIQA PROMPT TEMPLATE

Description: The model is tasked with selecting the most plausible solution to a given physical situation and must return its selection strictly in JSON format.

Prompt Text:

```
Task
Given the following situation, which option is more likely to be correct?
Situation: {entry['prompt']}
Option 0: {entry['solution0']}
Option 1: {entry['solution1']}

Output format
Return a valid JSON dictionary with the following key: 'answer' and a value should b
```

B COPA PROMPT TEMPLATE

Description: This template includes conditional phrasing based on whether the specific dataset entry requires identifying a cause or a result, as well as whether the target language matches the instruction language.

Prompt Text:

```
You will be given a task. The task definition is in English, [Condition A]. Here is
Given the premise "{entry['premise']}", and that we are looking for the [Condition B
Hypothesis 1: "{entry['choice1']}".
Hypothesis 2: "{entry['choice2']}".

Output format
Return a valid JSON dictionary with the following key: 'answer' and a value should b
```

Conditional Logic Notes:

- **[Condition A]:** If the evaluated task is in English, the prompt inserts "as is the task itself". If the task is in another language (e.g., Slovenian, Croatian), it inserts "but the task itself is in another language".
- **[Condition B]:** Inserts either "cause" or "result" depending on the specific relational requirement of the target premise.

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